

[Question 1] What is the primary goal of Artificial Intelligence?

- A. Replace human work
- B. To Create machines that exhibit human-like intelligence
- C. To build faster computers
- D. To automate manual tasks

[Question 2] What is the main goal of supervised learning?

-> To predict outcomes based on input data

[Question 3] What is a key feature of clustering algorithm?

-> They find natural groupings in data

[Question 4] Which library is commonly used for machine learning in Python?

- A. Pandas
- B. Numpy
- C. Scikit-learn

[Question 5] What is the main purpose of data processing?

-> To clean and prepare data for analysis.

[Question 6] Why it is important to have a large sample size?

- A. Increase variability
- B. Reduce sampling error

[Question 7] The principle component are

-> Orthogonal

[Question 8] What is the potential drawback of cross validation?

- C. It can be computationally expensive
- D. Increase data leakage

[Question 9] What is algorithm bias?

-> A systematic error in AI output

注意这个bias这里不应当翻译为偏置，而是单纯的算法偏见

[Question 10] What is relationship between model complexity and overfitting?

-> Increased complexity can lead to overfitting

[Question 11] In SVM, what is support vector?

- A. A data point far from decision boundary
- B. A data point close to the decision boundary
- C. Center of a data cluster
- D. A point that misclassified

[Question 12] What is the role of the kernel in SVM?

- A. Measure accuracy
- B. To transform data into higher-dimension space
- C. Reduce number of feature
- D. Optimal learning rate

[Question 13] What does a decision tree use to split data at each node?

- A. Random decision
- B. Splitting criteria like Gini impurity or entropy
- C. Data clustering
- D. Data normalization

[Question 14] What is the primary goal of Random Forest?

- A. To reduce dimension
- B. To improve predict accuracy by combining multiple trees

- C. Improve data size
- D. Cluster data points

[Question 15] Which of the following is a disadvantage of ensemble learning?

- A. Increased model complexity
- B. Reduce accuracy
- C. Less robust predict
- D. Simplified implementation

[Question 16] How is the distance between clustered in hierarchical clustering?

- A. Using centroids only
- B. Using linkage criteria
- C. Using decision tree
- D. Regression analysis

[Question 17] How does K-Means handle outliers?

- A. Ignore
- B. It is sensitive to them
- C. Remove automatically
- D. Cluster them separately

[Question 18] In GMM what does the weight of a component represent?

- A. The influence of each feature
- B. The probability of a data point belonging to the component
- C. The relative frequency of each Gaussian component in the data
- D. The distance between clusters